## Investigation: RECRUSUL S.A. (CVM code 12572), fiscal year 2012, as it was on 2024-12-31

Statistical signal based on public financial statements. Not an accusation, not investment advice.

## Company

Sector: Máquinas, Equipamentos, Veículos e Peças (origin: FCA); group: non-financial.
Before 2018, the company was called RECRUSUL SA.
Trading codes (2023 FCA, version 1, filed on 2023-05-08): RCSL3 (Ações Ordinárias, Bolsa, since 1985-12-20); RCSL4 (Ações Preferenciais, Bolsa, since 1985-12-20).

## Document used

2012 DFP, version 2, filed on 2013-04-04; origin: open data; type: consolidated; original scale: MIL.
Versions known on the date: 1 (2013-03-27); 2 (2013-04-04).
Document at the CVM: http://www.rad.cvm.gov.br/ENETCONSULTA/frmDownloadDocumento.aspx?CodigoInstituicao=1&NumeroSequencialDocumento=25920

## Possible manipulation (engine v1, study 1)

Beneish M-score
- Value: −1.207 (index)
- Alert by the study 1 rule: yes
- Rule: M > −1.78 (pre-registration, section 4)
- Percentile in sector and year: 91 (n = 17)
- Percentile in group and year: 91 (n = 364)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

DSRI, days' sales in receivables index
- Value: 1.596 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 98 (n = 21)
- Percentile in group and year: 92 (n = 457)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

GMI, gross margin index
- Value: 0.664 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 2 (n = 21)
- Percentile in group and year: 8 (n = 471)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

AQI, asset quality index
- Value: 0.959 (index)
- M-score component; it has no rule of its own
- Percentile in sector and year: 36 (n = 21)
- Percentile in group and year: 31 (n = 468)
- The company's own history: fewer than 5 previous fiscal years with a value
- Confidence in the data: medium

SGI, sales growth index
- Value: 1.491 (index)
- M-score component; it has no rule of its own

Percentile in sector and year: 88 (n = 21)
Percentile in group and year: 89 (n = 472)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DEPI, depreciation index
Value: 0.971 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 50 (n = 17)
Percentile in group and year: 49 (n = 383)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

SGAI, SG&A expenses index
Value: 0.351 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 2 (n = 21)
Percentile in group and year: 2 (n = 465)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

TATA, total accruals to total assets
Value: 0.071 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 98 (n = 22)
Percentile in group and year: 90 (n = 487)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

LVGI, leverage index
Value: 0.880 (index)
M-score component; it has no rule of its own
Percentile in sector and year: 17 (n = 21)
Percentile in group and year: 17 (n = 480)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

DPO jump
Value: −127.8 days
Alert by the study 1 rule: no
Rule: DPO change in the year's top decile among non-financial companies (≥ 90th percentile of computable values on the date: 32.5 days)
Percentile in sector and year: 2 (n = 20)
Percentile in group and year: 4 (n = 440)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Bargaining power with suppliers and a change in purchase mix (decision of 2026-11-07)

Implied interest rate minus average CDI
Value: 90.77 p.p.
Alert by the study 1 rule: yes
Rule: implied interest rate above the year's average CDI rate + 10 p.p. (pre-registration, section 4)
Percentile in sector and year: 89 (n = 22)
Percentile in group and year: 95 (n = 439)

The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium
Alternative explanations: Expensive debt, exchange differences and lease interest inside finance costs (decision of 2026-11-07)

Accruals to average total assets
Value: 0.077 (index)
Alert by the study 1 rule: does not trigger alerts
Percentile in sector and year: 98 (n = 22)
Percentile in group and year: 91 (n = 486)
The company's own history: fewer than 5 previous fiscal years with a value
Confidence in the data: medium

## Financial risk (context only)

Altman Z" (financial risk, context only)
Value: −6.803 (index)
Components: X1 = −0.290; X2 = −1.265; X3 = −0.072; X4 = −0.278 (indices)
Percentile in sector and year: 15 (n = 23)
The company's own history: fewer than 5 previous fiscal years with a value
- No cut-off zone: the available source does not define zones for Z"
- Coefficients estimated outside Brazil; bankruptcy models lose validity over time (Pereira and Martins 2015, p. 25)

## Financial quality (context only)

Piotroski F (financial quality, context only)
Value: F = 4/9 (8 computable)
Signals:
- Return on assets (ROA): 0
- Operating cash flow on assets (CFO): 0
- Change in ROA (ΔROA): 1
- Cash above earnings (ACCRUAL): 0
- Change in leverage (ΔLEVER): 0
- Change in current ratio (ΔLIQUID): 1
- No equity issuance (EQ_OFFER): —
- Change in gross margin (ΔMARGIN): 1
- Change in asset turnover (ΔTURN): 1
- No equity issuance (EQ_OFFER), from the cash flow statement: 6.03 lines of the frozen "emissões" pattern (reconnaissance, part 4.2), only those about shares or capital, as the article defines it ("did not issue common equity", p. 9); adaptation: preferred shares also count, because in Brazil they are equity, and the line is identified by text in the cash flow statement

## Changes

Restatement through the following year's comparative: no difference above the tolerance, or the next year's DFP did not exist yet on the date.

## What was known on the date

- DFP 2010, v1: 2011-03-29
- DFP 2011, v1: 2012-03-30
- FCA 2012, v1: 2012-03-30
- FCA 2012, v2: 2012-05-03
- FCA 2012, v3: 2012-12-14
- DFP 2012, v1: 2013-03-27
- DFP 2012, v2: 2013-04-04

- DFP 2013, v1: 2014-03-31
DFP and FCA filed with the CVM by the reference date; ITR and material facts come in later phases.

## Data quality
53 applicable checks; 0 failed.

## Auditor's report and auditor (factual section; no alert)
Auditor's report: Sem Ressalva (unqualified).
no opinion excerpt was found in the text
Auditor in the 2021 FCA: Taticca Auditores Independentes SS.
Auditor in the 2022 FCA: Taticca Auditores Independentes SS.
Auditor in the 2023 FCA: Taticca Auditores Independentes SS.
Auditor changes (one line per change, canonical name by the firm's CVM code):
- 2012: DRS Auditores -> Michelon & Cia Auditores e Consultores SS. The outgoing firm audited 5 consecutive fiscal years (2007 to 2011), by the start and end dates in the FCA. The change coincides with the end of the 5-fiscal-year term and is compatible with the rotation rule (art. 31 of CVM Instruction No. 308/1999, in force when the new firm started).
- 2015: Michelon & Cia Auditores e Consultores SS -> Baker Tilly Brasil RS Auditores Independentes Sociedade Simples. The outgoing firm audited 3 consecutive fiscal years (2012 to 2014), by the start and end dates in the FCA.
- 2020: Baker Tilly Brasil RS Auditores Independentes Sociedade Simples -> Taticca Auditores Independentes SS. The outgoing firm audited 5 consecutive fiscal years (2015 to 2019), by the start and end dates in the FCA. The change coincides with the end of the 5-fiscal-year term and is compatible with the rotation rule (art. 31 of CVM Instruction No. 308/1999, in force when the new firm started).

Auditor rotation: under CVM Resolution No. 23/2021, art. 31, the same independent auditor may not serve the same client for more than five consecutive fiscal years, with a minimum interval of three fiscal years before being hired again; under art. 31-A, the term is up to ten consecutive fiscal years when the company has a statutory audit committee in permanent operation and the auditor is a firm. The Resolution has been in force since 2021-04-01 (art. 41); before that, the equivalent rule was in arts. 31 and 31-A of CVM Instruction No. 308/1999, revoked by the Resolution (art. 40, I). A change of auditor is a fact, not an alert.

## How the engine was tested
In study 1, engine v1 detected 7 of 13 detectable cases, with 108.3 false alarms per detected case over the whole period and an alert rate of 49.9% among evaluated non-financial company-years (docs/summary.md).

## Limits
- No alert is not a clean bill of health. In Dechow, Sloan and Sweeney (1995), the most powerful accruals model rejected 'no manipulation' for only 28.1% of 32 companies sanctioned by the SEC (Table 7, p. 32).
- In study 1, 49.9% of evaluated non-financial company-years (48.3% including financial companies) were in alert, and the number of detections did not beat chance in a descriptive test done after the result: 7 detections in 13 detectable cases, against 7.37 expected by chance without assuming independent years (8.56 assuming independent years).
- Risk models use numbers that may be manipulated (Pereira and Martins 2015, p. 29). That is why the three families are not added up.

## Limitations
- Context indicators (Altman Z'' and Piotroski F) were not validated in Brazil by this project and do not trigger alerts

Evidence in the JSON: 38 DFP lines, each with document, version, filing date, account and location.

## Engine

motor v1 (estudo 1), branch estudo-1-final, commit 8c9a823; chiocca.v2 (produto, fases B a E).
Generated at 2026-11-11T00:49:34+00:00.

## Today's information (outside the reference date)

Registry status: ATIVO. Source: CVM registry of listed companies (cad_cia_aberta.csv), in the version downloaded for the project, not on the reference date.
Name in today's registry: RECRUSUL SA.